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Calculus examples and function



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Understanding Calculus

Key Concepts and Role in Machine Learning

What is Calculus?

The study of continuous change in mathematics

Main Branches:

- Differential Calculus: Focuses on change rates
- Integral Calculus: Deals with accumulation

Why It Matters in ML?

Used for optimisation, backpropagation, and function analysis

Essential Calculus for ML

Core Concepts You Need to Know

1

Derivatives and Gradients

2

Partial Derivatives

3

Gradient Descent Optimisation

4

Hessian Matrix and Second Derivatives

5

Integration for Probability Distributions

6

Jacobian and Hessian in Neural Networks

Understanding Derivatives & Gradients

Key Takeaways

Definition: Represents how a function $f(x)$ changes concerning x

Example:

If **$(f(x) = x^2)$** , then **$(f'(x) = 2x)$** , meaning the rate of change increases as x increases

Applications in ML:

- Helps optimise cost functions
- Fine-tunes model parameters to minimise errors

Partial Derivatives in ML

Understanding Variable Sensitivity in Functions

Definition: The partial derivative of $f(x, y)$ with respect to x keeps y as a constant

Example:

If $(f(x, y) = x^2 + y^3)$, then $(\partial f / \partial x = 2x)$ and $(\partial f / \partial y = 3y^2)$

Applications in ML:

- Essential for reducing dimensionality
- Useful for improving computational efficiency

Hessian Matrix & Its Role in ML

Understanding Second-Order Derivatives in Optimisation

Definition: A square matrix of second-order partial derivatives

Formula:

$$[H(f) = [\partial^2 f / \partial x^2 \quad \partial^2 f / \partial x \partial y; \quad \partial^2 f / \partial y \partial x \quad \partial^2 f / \partial y^2]]$$

Applications in ML:

- Analyse curvature and convergence
- Apply in second-order optimisation techniques

Integration in Probability

Calculating Areas for Probabilistic Insights

Definition:

- Integration calculates the area under a function's curve

Application in ML:

- Use in probability distributions (e.g., Normal distribution)
- Utilise in Bayesian inference and expectation calculations

Example:

- Compute cumulative probabilities from a probability density function (PDF)

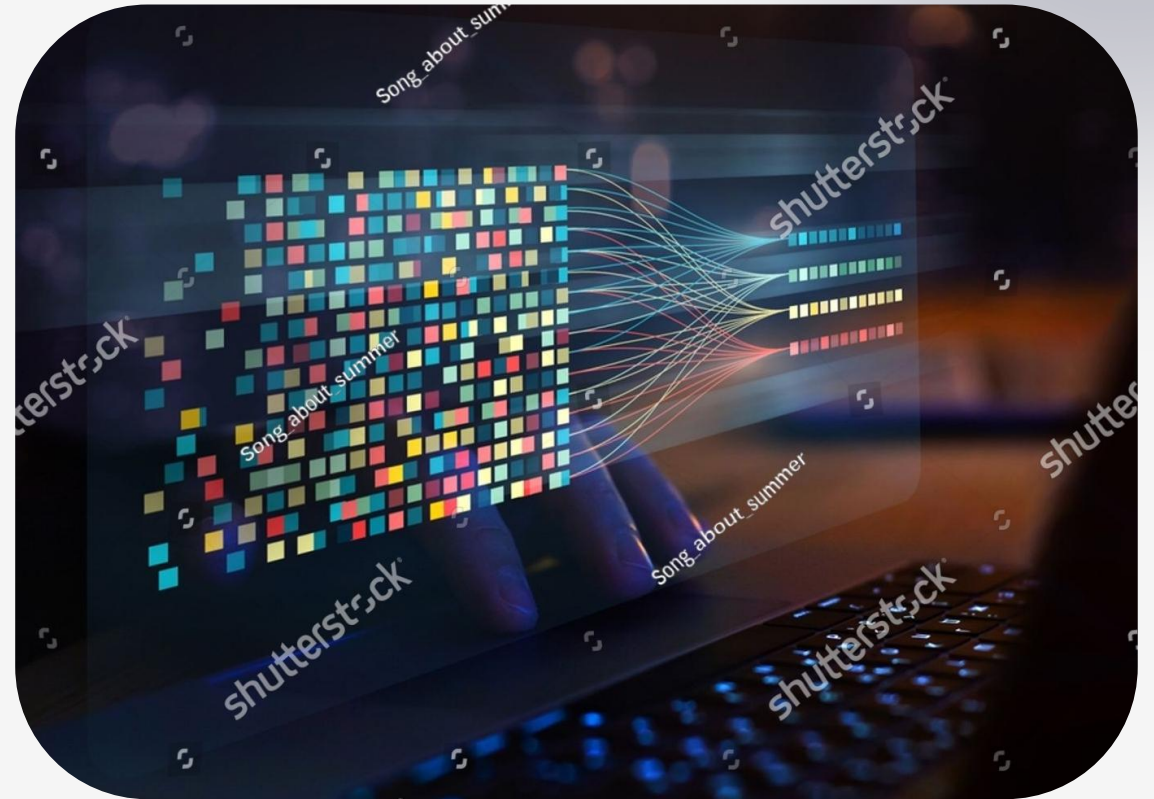
Jacobian & Hessian in Neural Networks

Understanding Derivatives in Optimisation

- **Jacobian:** Matrix of first-order partial derivatives
- **Hessian:** Matrix of second-order derivatives (measures curvature)

Application in ML:

- Utilising integration in deep learning enhances optimisation
- Aiding second-order methods like Newton's method, integration improves efficiency



Differentiation in Machine Learning



1

Differentiation measures a function's rate of change

2

crucial for optimisation, backpropagation, and regularisation in ML

3

Minimises cost functions and enhances model performance

Gradient Descent

Optimisation Algorithm in ML

Definition:

- An optimisation algorithm iteratively updates model parameters by following the cost function's gradient

Formula:

$$[\theta = \theta - \alpha \partial J(\theta) / \partial \theta]$$

where (α) is the learning rate

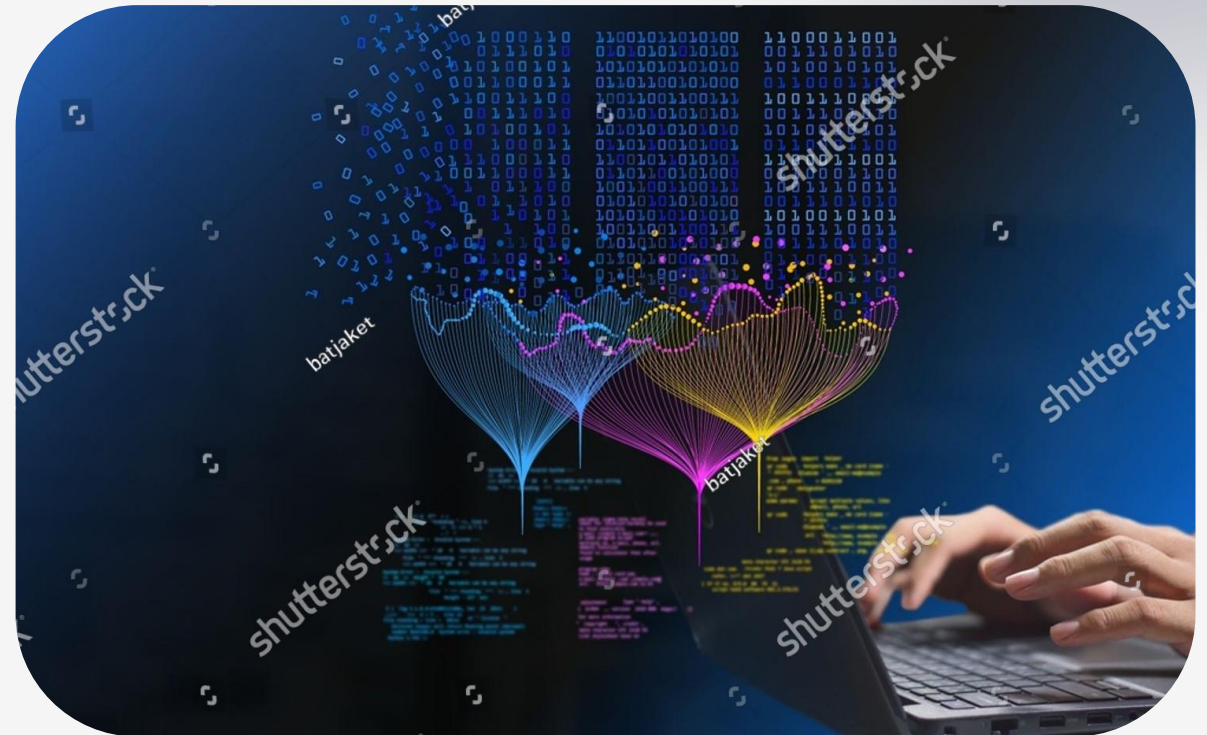
Application:

- Used in training machine learning models, including deep learning

Finding the Minimum of a Cost Function

Optimisation Using Gradient Descent

- **Quadratic cost function:** $J(\theta) = \theta^2$
- **Derivative:** $dJ/d\theta = 2\theta$
- **Gradient Descent Update:** Adjust θ iteratively
- **Converges to Minimum:** $\theta = 0$



Backpropagation in Neural Networks

Optimising Weights Using Gradients



1 Forward pass: Compute predictions

2 Loss calculation: Measure error

3 Gradient computation: Apply chain rule

4 Weight update: Use gradient descent

Backpropagation optimises neural networks using differentiation.

Chain Rule in Deep Learning

Fundamental Rule for Backpropagation

Functions:

$$(z = f(y)) \text{ and } (y = g(x))$$

Chain Rule:

$$[dz/dx = dz/dy \times dy/dx]$$

Applied in deep learning to propagate error backward

Derivatives of Activation Functions

Key derivatives of activation functions used in neural networks

Function	Definition	Derivative
ReLU	$(f(x) = \max(0, x))$	1 for $(x > 0)$, 0 otherwise
Sigmoid	$(f(x) = 1/(1 + e^{(-x)}))$	$[f'(x) = f(x) (1 - f(x))]$
Softmax	Utilises for classification	Computes gradients in backpropagation

Regularisation Techniques

Prevents Overfitting

Purpose: Adds a penalty to the loss function

L1 Regularisation (Lasso):

- Adds $(\lambda \sum |\theta|)$ to the loss
- Promotes sparsity
- Derivative: Sign function

L2 Regularisation (Ridge):

- Adds $(\lambda \sum \theta^2)$ to the loss
- Promotes small weights
- Derivative: $(2\lambda\theta)$

ML Optimisation

Minimising Cost Functions



- Optimisation reduces cost functions efficiently
- Differentiation-based techniques drive ML algorithms

Unconstrained Optimisation

Minimising Cost Functions

Example: Logistic Regression Cost Function

$$[J(\theta) = - 1/m \sum y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Method: Uses gradient descent to minimise error

Constrained Optimisation (Lagrange Multipliers)

Definition & Example

Definition: Optimisation with constraints

Example: Support Vector Machines (SVM)

- Maximises the margin with a constraint on misclassification
- Uses Lagrange multipliers to enforce constraints

Convex vs. Non-Convex Functions

Understanding Optimisation in ML

- **Convex Functions:** One global minimum (e.g., Mean Squared Error, Cross-Entropy Loss)
- **Non-Convex Functions:** Multiple local minima (e.g., deep learning loss)
- Convexity ensures optimisation convergence

Functions in Machine Learning

Basics and Applications

Definition: A function maps input (x) to output (y)

Role in ML: Defines relationships, transformations, and optimisations

Applications: Used in activation, loss functions, and feature transformations



Functions in Machine Learning

Common Types

1

Linear

2

Quadratic

3

Exponential

4

Logarithmic

5

Sigmoid

6

ReLU (Rectified Linear Unit)

Linear Functions

Definition, Graph, and Applications

Definition:

- A function of the form

$$(f(x) = ax + b)$$

Graph: A straight line

Applications in ML:

- Use in linear regression
- Form the foundation of many ML models
- Help in feature scaling and transformation

Quadratic Functions in Machine Learning

Key Concepts and Applications

Definition:

- A function of the form

$$(f(x) = ax^2 + bx + c)$$

Graph: A parabola

Applications in ML:

- Using in optimisation problems
- Mean Squared Error (MSE) is a quadratic loss function
- Applying in feature engineering transformations

Exponential Functions in Machine Learning

Key Concepts and Applications

Definition:

- A function of the form

$$(f(x) = a^x)$$

Graph: Shows rapid growth or decay

Applications in ML:

- Using in learning rate decay for optimisation
- Models growth patterns like population growth and compound interest
- Finding in probability distributions (e.g., exponential distribution)

Logarithmic Functions in Machine Learning

Key Concepts and Applications

Definition:

- A function of the form

$$(f(x) = \log(x))$$

Graph: Gradually increasing function

Applications in ML:

- Using in log transformations for normalising skewed data
- Helping in feature scaling during preprocessing
- Finding in loss functions like log loss (cross-entropy)

Sigmoid Function in Machine Learning

Key Concepts and Applications

Definition:

- A function of the form

$$(f(x) = 1 / (1 + e^{-x}))$$

Graph: S-shaped curve

Applications in ML:

- Using as an activation function in neural networks
- Converting values into probabilities (0 to 1)
- Using in logistic regression

ReLU (Rectified Linear Unit) Function

Key Concepts and Applications

Definition:

$$(f(x) = \max(0, x))$$

Graph: Linear for positive values, zero for negatives

Applications in ML:

- Most commonly used as an activation function in deep learning
- Helps overcome the vanishing gradient problem
- Enables faster training in neural networks

Activation Functions in Neural Networks

Purpose, Types, and Importance

Purpose: Introduce non-linearity in neural networks

Types: Sigmoid, ReLU, Tanh, Leaky ReLU

Importance: Helps deep learning models learn complex patterns



Loss Functions in Machine Learning

Understanding, Key Types, and Significance

Understanding: Loss functions quantify prediction errors

Key Types: MSE (Quadratic), Cross-Entropy (Logarithmic), Hinge Loss (SVMs)

Significance: Crucial for model optimisation and training



Feature Transformation in Machine Learning

Purpose, Types, and Benefits

Purpose: Enhances model performance

Types: Log transformation (logarithmic), polynomial features (quadratic), normalisation and scaling (linear)

Benefits: Improves interpretability and efficiency



Q&A Session



Open floor for questions and clarifications.



Discussion on project groups and next sessions.